Welcome to The AMOS Demo Day!

Dirk Riehle, FAU Erlangen

AMOS C02

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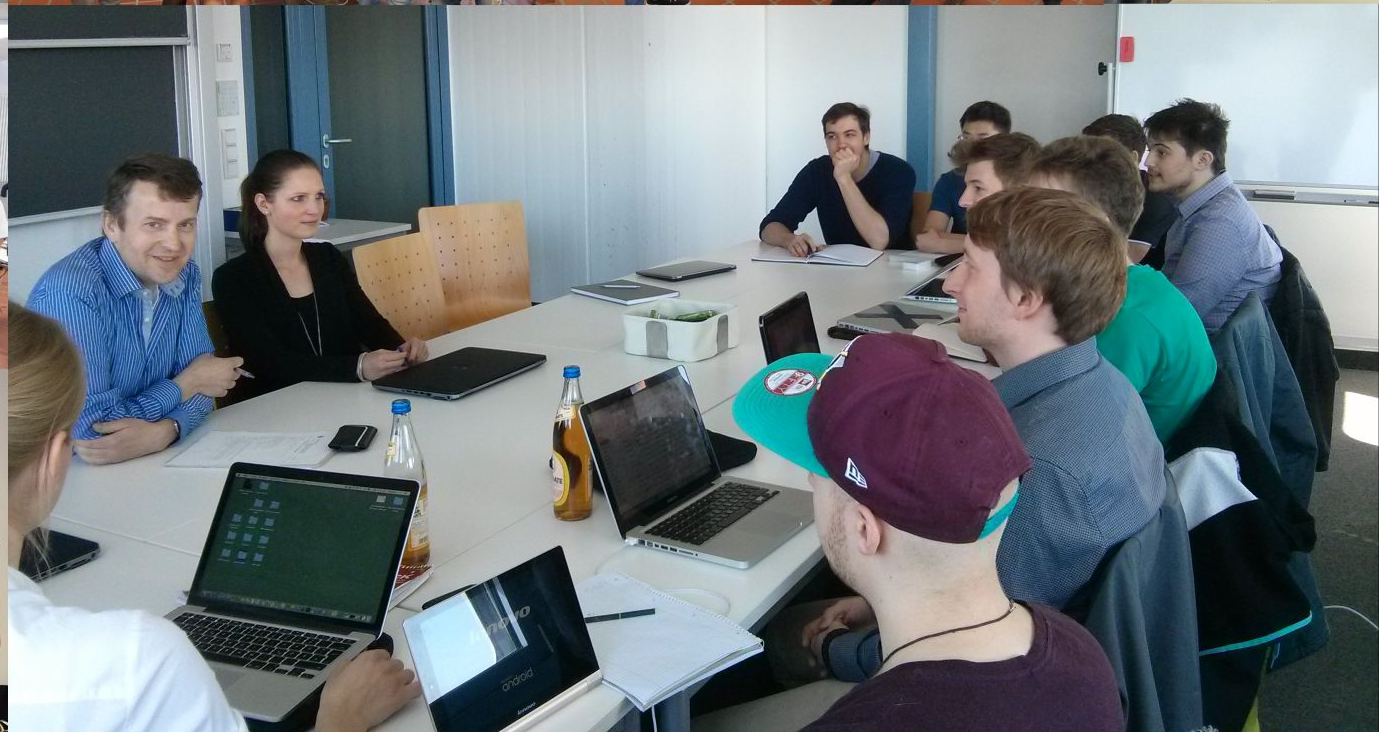
The Professorship of Open Source Software



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Industry Partners



Participating Universities



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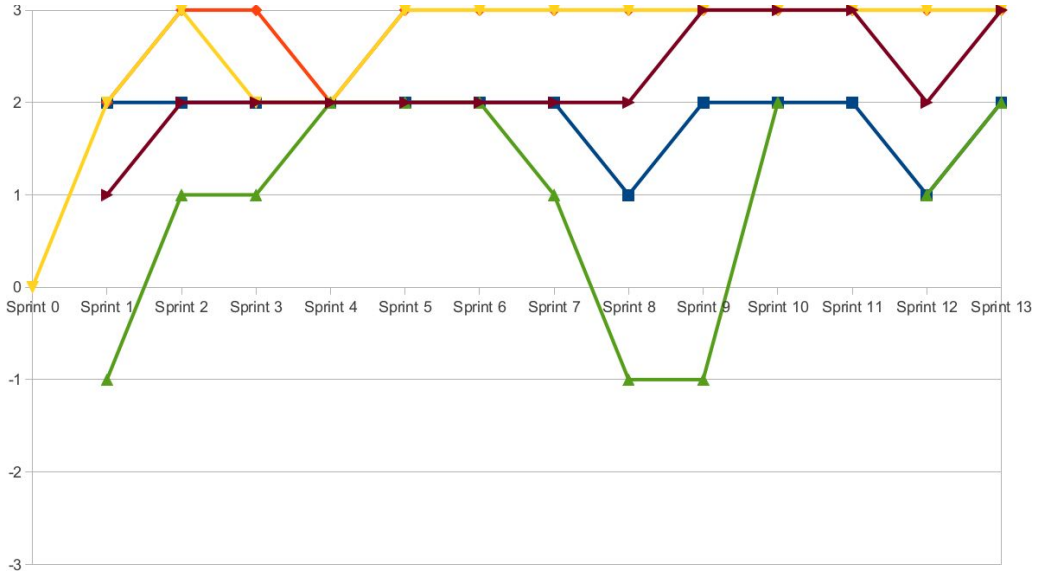
The AMOS Scrum Team

DR



Project Coaching and Support





Demo Day Schedule

Time	Duration	Responsible	Title	Room
10:15	10 min	Riehle	Introduction	Main room
10:25	10 min	Teams	One slide summary	Main room
10:35	20 min	Teams	Demo	Demo rooms
10:55	20 min	Teams	Demo	Demo rooms
11:15	20 min	Teams	Demo	Demo rooms
11:35	5 min	Riehle	Conclusions	Main room

This Semester’s Projects

1. Zero Downtime Linux Updates with Weber
2. Automated Thematic Analysis with N.I.M.
3. Allixir Intelligence with Zink (startup)
4. TaskOrbit Conversational Agent with Meisterwerk

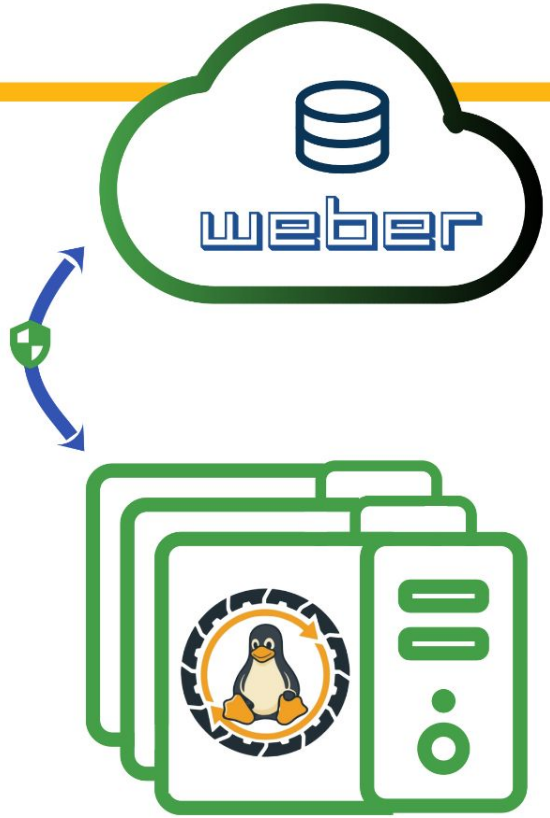
Zero-Downtime Linux Updates

Simple. Scalable. Secure.

Problem: Industrial edge ipc fleets can’t afford production stops for routine OS updates.

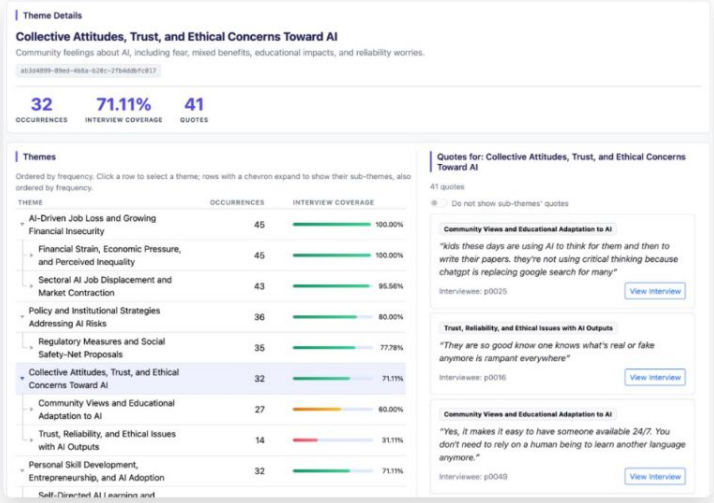
Solution: Container-based atomic OS swap with automatic fallback on failure. Secured by TPM.

Live Demo: Watch an OS update roll out to an edge device with near zero downtime!



Group 2: Automated Thematic Analysis

- An interview analysis tool for social sciences researchers.
- Automatically extract recurring themes from interviews using state-of-the-art AI.
- Generate insights based on participant demographic data.



AMOS SS 2026 Project

FUNDED BY
Nuremberg Institute for Market Decisions (NIM)
nim.org →

ACADEMIC PARTNER
Professorship for Open-Source Software
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Team: Alexander Schmidt, Christoph Mantsch, Fabiano de Sa Filho, Hamid Rasul, Julian Wendlandt, Mohammad Annus, Ole Müermann, Paul Ammann, Paul Klarer, Selin Kahvecioglu

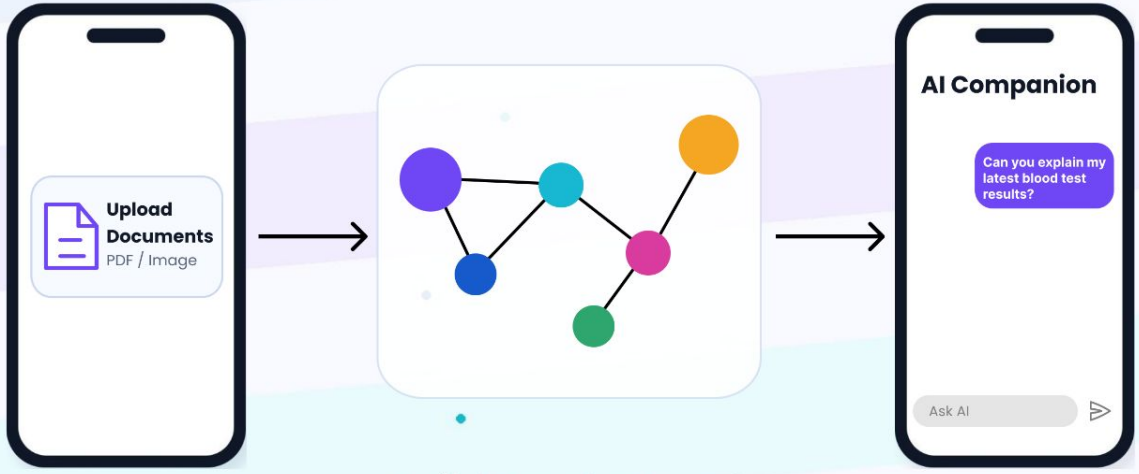


Ailixir Intelligence

Distilling Knowledge into Actionable Insights

Problem:

- Information scattered across **time** and **many documents**
- **Incomprehensible** Jargon (e.g. medical reports)
- Regular AI chat bots **lose context** and **hallucinate** with too much input



Build Graph-Based Knowledge Base

Friedrich-Alexander-Universität Erlangen-Nürnberg

TaskOrbit

VOICE WORKFLOW ORCHESTRATION

TaskOrbit turns individual scenario agents into deployable, end-to-end voice multi-agent workflows

VALUE PROPOSITION

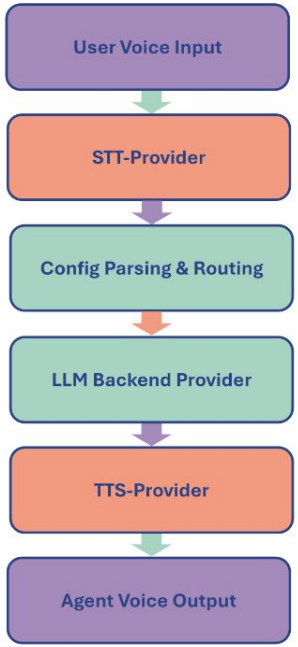
A technology foundation for company-specific processes — not a one-size-fits-all solution.

- Build and deploy reusable scenario agents
- Compose agents into cross-scenario workflows
- Automate interactions from speech input to spoken response

PROJECT SCOPE

- Agent runtime and workflow execution
- Benchmarking for custom workflows
- Basic tools: end calls, forwarding and external data references
- Flexible inference: External, self-hosted open-source models or OpenRouter

CORE PIPELINE



TypeScript React Vite Python Node.js

ElevenLabs Deepgram LiveKit GCP OpenRouter

Let’s go see the demos!

Please join a breakout session [1]

[1] If you don’t see a choice window, please look to Breakout rooms in the menu bar or under More

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